Simulation of ground truth interaction networks from microbial community model

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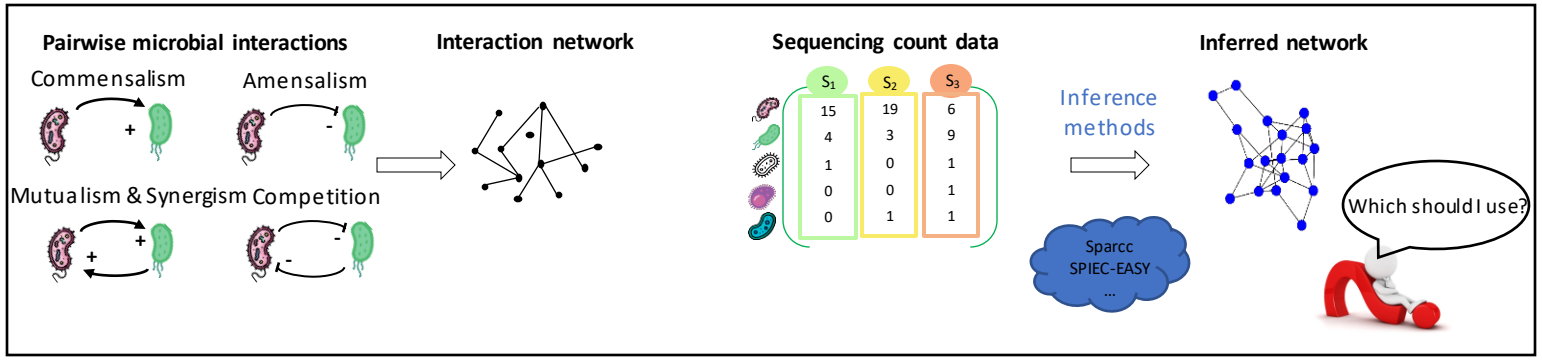
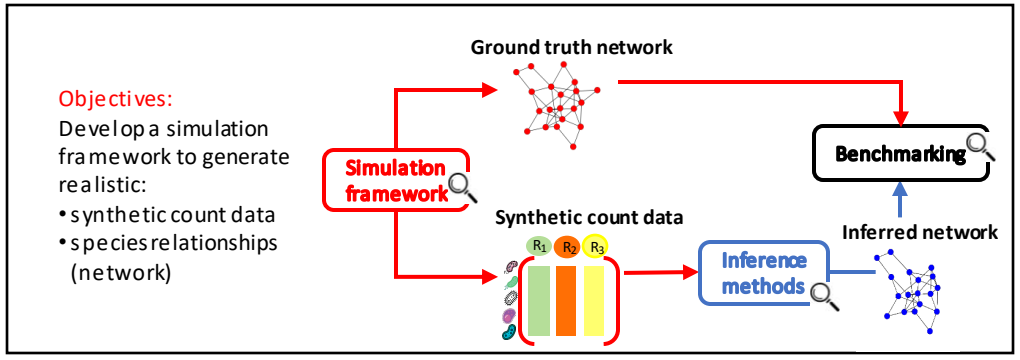
RECONSTRUCT MICROBIAL INTERACTION NETWORK

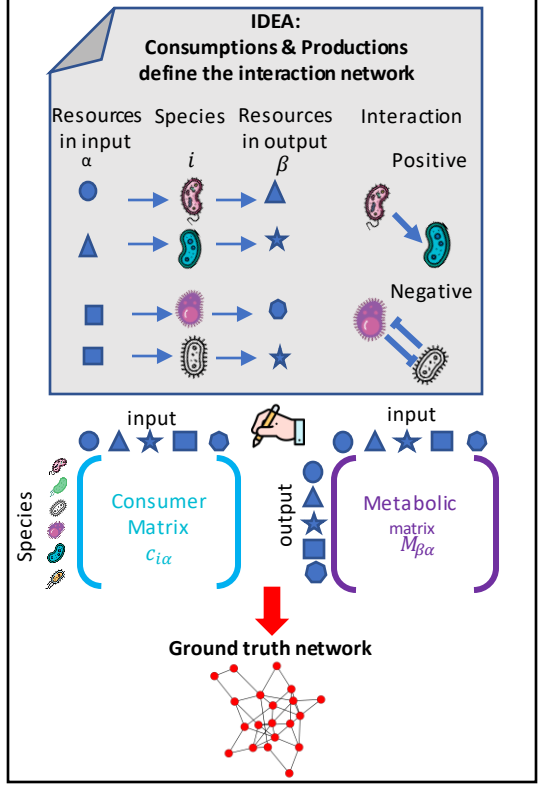


HOW TO CHOOSE ?

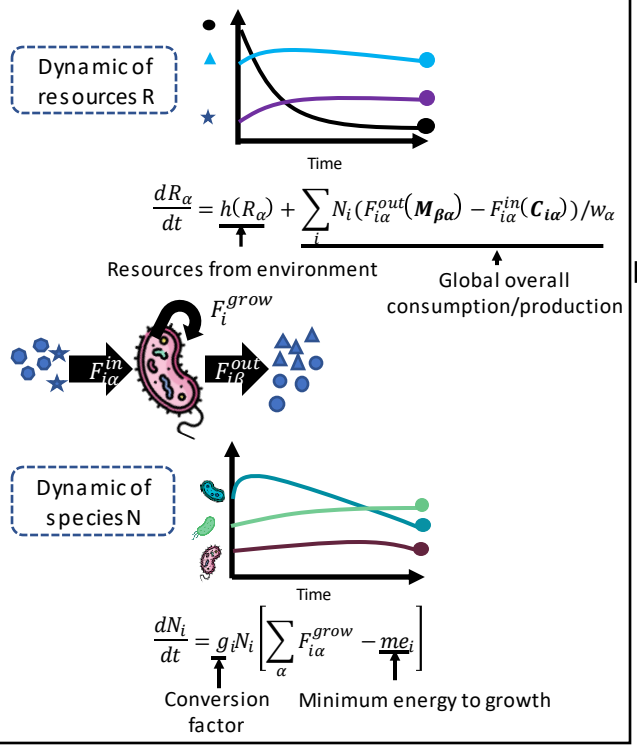


Simulation framework

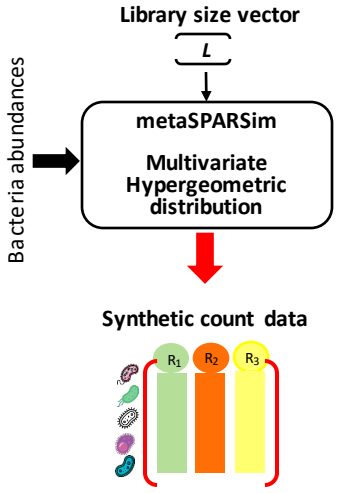
1) Model interspecies interactions based on metabolic consumption/production



2) Simulate microbial population dynamics using " Community Simulator" (Marsland et al., PLoS ONE, 2020)



3) Simulate sequencing process using "metaSPARSim" (Patuzzi et al., BMC Bioinformatics, 2019)



Inference methods

- Pearson's correlation
- Partial Pearson's correlation
- Spearman's correlation
- Partial Spearman's correlation
- Sparcc
- MIC
- SPIEC-EASY
- Ccrepe

Benchmarking

Network topology metrics
- Clustering coefficients
- Betweenness centrality
- Node degree distribution

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Edge recovery metrics
- AUPRC
- AUROC
- MSE

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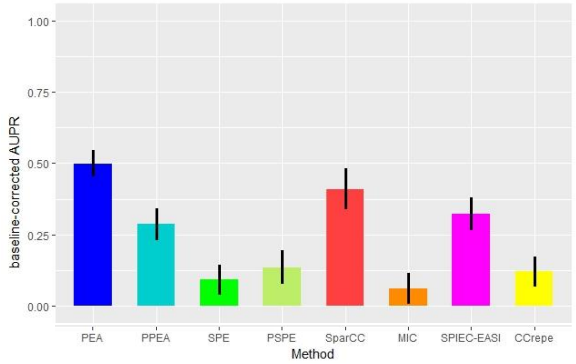
PRELIMINARY RESULTS